

New Records of Reef Fishes from the Andaman and Nicobar Islands

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INTRODUCTION

Studies on ichthyofauna of the Andaman and Nicobar Islands (ANI) are known from the earlier works of Talwar (1990), Devi (1991), and subsequent reports by several researchers (Rao *et al.*, 1992a,b, 1993a,b, 1994, 2000; Krishnan and Mishra, 1992a,b; Devi *et al.*, 1993; Dhandapani and Mishra, 1993; Rajan *et al.* 1993; Rao, 2003). Rao (2009) reported a checklist of fishes with 1369 species under 586 genera belonging to 175 families recorded from ANI, based on field survey and available literature. Dam Roy *et al.* (2009) added 11 more species to the ichthyofauna of these islands. Recently, Ramakrishna *et al.* (2010) reported 83 new records with an updated checklist containing 1463 species from these Islands.

During the recent survey on taxonomic studies of fishes carried out using underwater photography in Mahatma Gandhi Marine National Park (MGMNP), nine new records of fishes from ANI were made. In this chapter a detailed description of morphological features of the newly recorded fish species with their habitat and distribution are presented.

METHODS

Undersea surveys were carried out by SCUBA diving to study fish from five sites in Mahatma Gandhi Marine National Park (MGMNP) at a maximum depth of 25 m from April 2011 to June 2012. Underwater photography was taken by Sony Cyber Shot (DSC-T900) camera with underwater housing facility. The fishes were identified based on the standard systematic taxonomic keys (Weber and De Beaufort, 1913–1953).



FIGURE 14.1 *Apogon neotes*.

RESULTS AND DISCUSSION

Systematic Account (Order Perciformes)

Family Apogonidae

1 *Apogon neotes* (Allen, Kuitert & Randall, 1994) (Figure 14.1)

Common Name Larval cardinalfish

Material Observed Tarmugli Island; depth 15–20 m.

Diagnostic Characters The specimen was approximately 3–4 cm. Body oblong and compressed; head large, snout short; eye big in size; caudal emarginated. Body silvery-blue, translucent; black stripe intermittently bordered with silver stripes runs from above eye to back spot on mid-tail base.

Distribution Indonesia, Palau, Philippines, Papua New Guinea, and Solomon Islands.

Habitat & Ecology Found around coral reefs at varied depths in range 15–30 m in clear waters of the outer reef and lagoons. It is typically seen in areas of Gorgonian fans and soft corals. Solitary, in pairs or loose clusters and using crevices in the reef for shelter. Feeds on zooplankton and small, bottom-dwelling crustaceans.

Remarks This species exhibits distinct pairing during courtship and spawning. The males are known for the unusual behavior of incubating egg masses inside their mouth.



FIGURE 14.2 *Monotaxis heterodon* (juvenile).

Family Lethrinidae

2 *Monotaxis heterodon* (Bleeker, 1854) (Figure 14.2)

Common Name Redfin Bream

Material Observed Grub Island and Tarmugli Island; depth 5–7 m.

Diagnostic Characters Attains 8–10 cm. Body oblong and compressed laterally; eyes large, constituting less than one third of head length. The operculum, cheek and inner surface of pectoral fin base are scaled. Body color is brownish on the dorsal surface gradating to pale underside with 3–4 narrow white bars; black spot in axil of pectoral fin; yellow tail lobes.

Distribution Seychelles to Marshall Islands and New Caledonia

Habitat & Ecology Tropical carnivorous fish. Found around coral reefs, lagoons and outer reef slopes up to 100 m.

Family Pomacentridae

3 *Pomacentrus pavo* (Bloch, 1787) (Figure 14.3)

Common Name Blue Damsel/ Sapphire Damsel.

Material Observed Chester Island, Grub Island, Jolly Buoy and Tarmugli Islands; depth 8–10 m.

Diagnostic Characters Small fish; attains 6–9 cm. Body oblong and slightly compressed laterally; head profile convex. Body light blue to light green with



FIGURE 14.3 *Pomacentrus pavo*.

vertical blue to grey streaks on each scale; scattered blue spots on head; dark 'ear' spot present; spinous dorsal pale with blue margin, remainder of fin pale to dusky.

Distribution East Africa to Micronesia and French Polynesia. Taiwan to E. Australia.

Habitat & Ecology Good aquarium fish seen in loose groups on sandy areas down to 15 m around coral reefs. Feeds on plankton, filamentous algae and small invertebrates.

4 *Pomacentrus Polyspinus* (Allen, 1991) (Figure 14.4)

Common Name Thai Damsel.

Material Observed Red Skin Island; depth 3–5 m.

Diagnostic Characters Attains 7–8 cm. Body compressed. Color greenish-grey with dark scales; small blue lines on snout and forehead; a blue-edged dark spot on rear dorsal fin; dark 'ear' spot present; anal fin margin blue; caudal and lobes of adjacent fins translucent.

Habitat & Ecology Found in shallow to moderate depth in coral reef areas. They usually swim around the reef in pairs. Feeds on algae and small benthic animals.

5 *Pomacentrus proteus* (Allen, 1991) (Figure 14.5)

Common Name Colombo Damsel.



FIGURE 14.4 *Pomacentrus polyspinus*.



FIGURE 14.5 *Pomacentrus proteus*.

Material Observed Tarmugli Island; depth 5–7 m.

Diagnostic Characters Attains 6–8 cm. Body yellowish brown with blue streaks on scales; a blue line from snout to origin of spinous dorsal; a blue-edged dark spot on rear dorsal fin; In juvenile, dorsal half blue gradating to yellow below; light blue lines marking on upper head; rear dorsal fin with blue edged dark spot.

Distribution Sri Lanka.



FIGURE 14.6 *Pteragogus cryptus*.

Habitat & Ecology They are encountered in coral reef flats at depth of 5–10 m. The juveniles are seen to stay inside the shelter offered by branching corals. Adults are usually found on silty shoreline on mixed coral rubble. Feeds on plankton, filamentous algae and small invertebrates.

Family: Labridae

6 *Pteragogus cryptus* (Randall, 1981) (Figure 14.6)

Common Name Cryptic Wrasse.

Material Observed Tarmugli Island; depth 20–25 m.

Diagnostic Characters Attains 10–15 cm. Head pointed; caudal fin round. Body mottled shades of brown-red with scattered dark brown spots; white line passes from snout above eye to caudal; an oval dark spot, edged in pale on gill cover.

Distribution Red Sea, E. Africa to Samoa, Philippines and Micronesia to Australia.

Habitat & Ecology Found solitary beyond 60 m; usually swim around the reefs, sea grass beds and algal flats.

Family: Gobiidae

7 *Eviota sigillata* (Jewett and Lachner, 1983) (Figure 14.7)

Common Name Seven-figure Pygmy Goby.



FIGURE 14.7 *Eviota sigillata*. (Please see color plate at the back of the book.)

Material Observed Tarmugli Island; depth 8–9 m.

Diagnostic Characters Very small fish; attains 2.5–3 cm. Head compressed; ventral separate; caudal fin rounded. Body translucent with white and brown dash marks alternate on spinal column; pectoral fin base white.

Distribution Seychelles to Papua New Guinea, Micronesia, Solomon Islands and Australia.

Habitat & Ecology Occur in sheltered areas of the coral reef at depths of 2–20 m. Mostly resting on live corals and on coral rock. Very small fish; because of its size and transparency is often difficult to notice.

8 *Koumansetta hectori* (Smith, 1957) (Figure 14.8)

Common Name Yellowstripe Goby.

Material Observed Tarmugli, Grub and Red Skin Islands; depth 10–12 m.

Diagnostic Characters Body bluish-black with 4 yellow longitudinal stripes from snout to back; 3 dark spots, one on first dorsal fin, yellow edged second spot on second dorsal fin, and third spot on upper tail base. Caudal fin pale.

Distribution Red Sea, Indonesia, Palau and S.W. Japan.

Habitat & Ecology Found in sand bottom or rubble close to the reef edge at depth of 3–30 m. Observed solitary or in pairs.

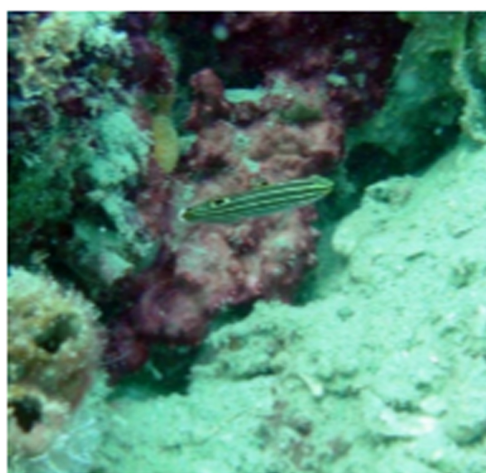


FIGURE 14.8 *Koumansetta hectori*. (Please see color plate at the back of the book.)

9 *Valenciennea limicola* Hoese & Larson, 1994 (Figure 14.9)

Common Name Mud Goby.

Material Observed Tarmugli Island; depth 5–8 m.

Diagnostic Characters Attains 9–10 cm. Snout elongate; ventral separate; caudal fin rounded. Body grey with pair of bright orange lines: one behind eye to caudal and other from snout to caudal; bright light-blue stripe below eye.



FIGURE 14.9 *Valenciennea limicola*. (Please see color plate at the back of the book.)

Distribution Gulf of Thailand and Indonesia to Fiji.

Habitat & Ecology Solitary or in pairs. Found around rubble on coral reefs or mud bottom of coastal reefs in 5–30 m.

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